

- **Electromagnetic:** 1.9683×10^{13} Hz (generated IR)
- **Blue Light:** $\sim 6.17 \times 10^{14}$ Hz (H-beta emission)
- **UV Light:** $> 10^{15}$ Hz (plasma emission)

Experimental Predictions

This model predicts:

1. **Spectral Analysis:** Should show hydrogen combustion lines, not just broad blackbody spectrum
2. **Timing:** Two-phase emission (blue combustion, then UV collapse)
3. **Bubble Growth:** Gradual bubble expansion over multiple cycles before flash
4. **Gas Detection:** H₂ and O₂ should be detectable in solution
5. **Frequency Dependence:** Effect should be strongest at acoustic frequencies that generate IR matching water vibrational modes

Advantages Over Traditional Model

Traditional Sonoluminescence: Purely thermal plasma from acoustic heating **This Model:**

- Explains specific blue spectral content
- Accounts for energy accumulation mechanism
- Provides electromagnetic pathway for energy conversion
- Explains two-phase light emission (blue + UV)
- Lower energy threshold than pure plasma formation

Conclusion

This electromagnetic-chemical model suggests that "sonoluminescence" is actually a complex process involving:

1. Acoustic → electromagnetic frequency conversion
2. IR-driven water splitting at bubble interface
3. Progressive gas accumulation via acoustic pumping
4. Hydrogen-oxygen combustion producing blue light
5. Post-combustion collapse generating UV emission

The phenomenon represents a novel form of acoustic-electromagnetic-chemical energy conversion rather than simple cavitation heating.